

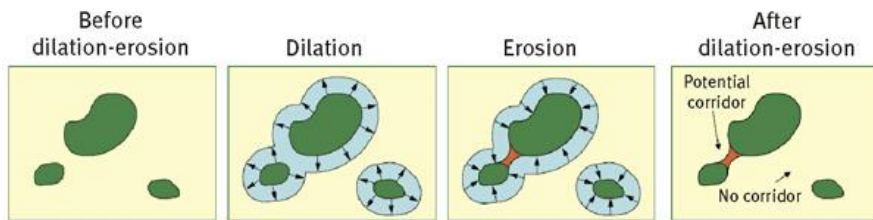
Mathematical Morphology

Lectures on Image Processing

Brno University of Technology, Faculty of Information Technology
Božetěchova 1/2, 612 66 Brno - Královo Pole



„Morphology is the study of size, shape and internal structure of objects “

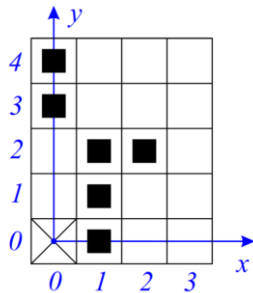


- Binary mathematical morphology
 - Dilation and erosion
 - Opening and closing
- Skeleton
- Distance transform
- Grayscale morphology

- Literature
 - Serra, J.: Image Analysis and Mathematical Morphology, Academic Press, London, 1988.
 - Najman, L. and Talbot, H.: Mathematical Morphology: From Theory to Applications, Wiley-ISTE, 2010

- Extracts knowledge from the relation of an image and a small structuring element, which is a predefined shape.
- The structuring element is checked in each pixel, how does this shape matches or misses local shapes in the image.
- Point set
- Structuring element
- Morphological transformation

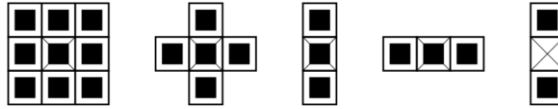
- Images can be represented by point sets of an arbitrary dimension



$$X = \{(1,0), (1,1), (1,2), (2,2), (0,3), (0,4)\}$$

- Binary mathematical morphology in 2D – a set of points represented as pairs of integers ($\in \mathbb{Z}^2$)
- Grayscale mathematical morphology in 2D – triplet ($\in \mathbb{Z}^3$)

- A structuring element serves as a local probe in morphological operators

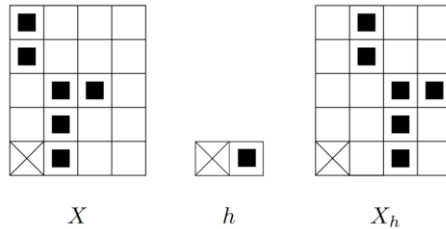


- Structuring elements are expressed with respect to local coordinates with a defined origin

- Relation between the image (point set X) and the typically smaller point set – the structural element

- E.g. translation of point set X by vector h

$$X_h = \{p \in \mathbb{E}^2, p = x + h \text{ for some } x \in X\}.$$



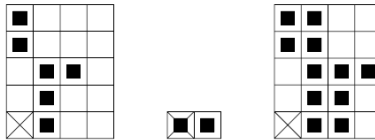
- Application of a morphological transformation to the image X corresponds to the systematic shifting of the element B in the image

- Binary dilation is (Minkowski) **addition of two point sets**:

$$X \oplus B = \{p \in \mathbb{E}^2 : p = x + b, x \in X \text{ and } b \in B\}$$

- i.e. union of translated point sets:*

$$X \oplus B = \bigcup_{b \in B} X_b.$$



$$X = \{(1,0), (1,1), (1,2), (2,2), (0,3), (0,4)\}$$

$$B = \{(0,0), (1,0)\}$$

$$X \oplus B = \{(1,0), (2,0), (1,1), (2,1), (1,2), (2,2), \dots\}$$

ABCD	ABCD
abcd	abcd
EFGHIJKLMN	EFGHIJKLMN
efghijklmn	efghijklmn

- Fills small holes and narrow bays in objects
- **The size of the object increases!**
 - If the size has to be kept similar then dilation is combined with erosion...

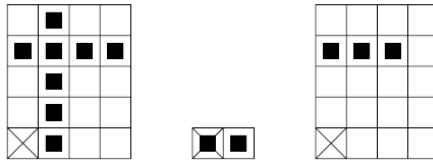
- (Minkowski) subtraction of two point sets:

$$X \ominus B = \{p \in \mathbb{E}^2 : p + b \in X \text{ pro každé } b \in B\}$$

- i.e. *intersection of all image X translations* by the vector $-b$:

$$X \ominus B = \bigcap_{b \in B} X_{-b}.$$

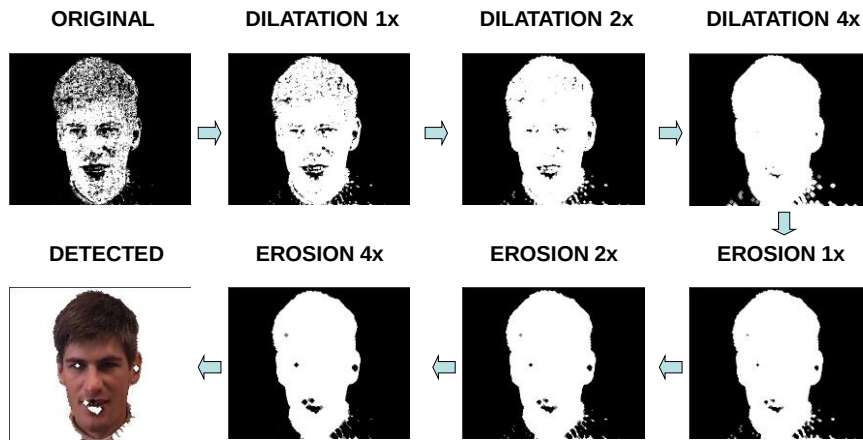
- For each image pixel p , it is verified whether the result fits to X for all possible $p+b$

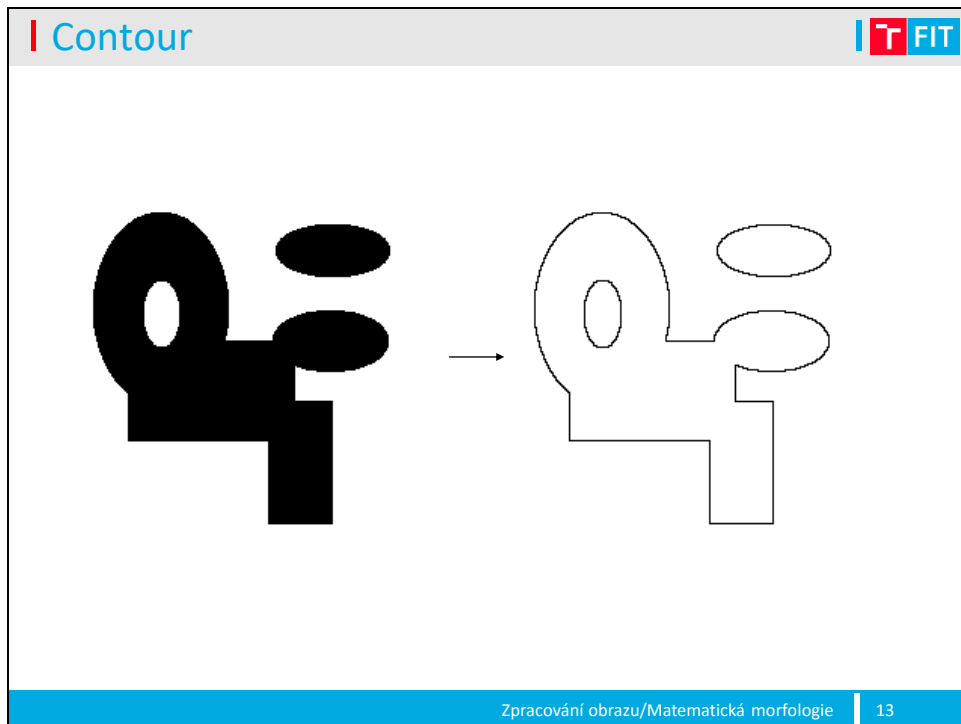


- Dilation and erosion are dual morphological operations

ABCD	ABCD
abcd	abcd
EFGHIJKLMN	EFGHIJKLLIN
efghijklmn	efghijklmn

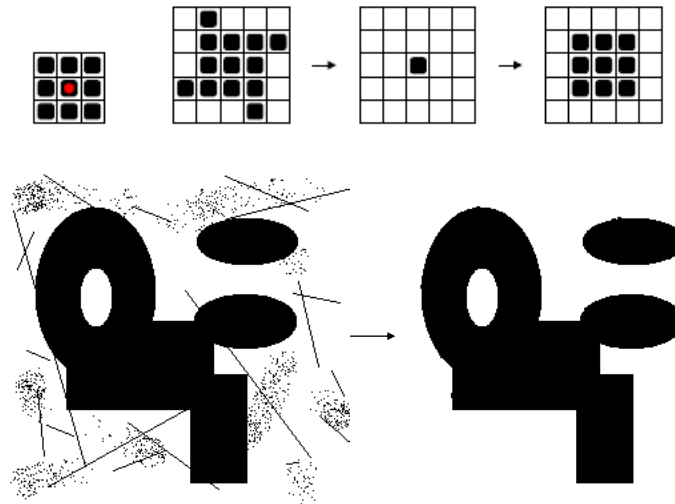
- Objects smaller than the structuring element vanish
- Erosion is used for simplifying structure (decomposition of an object into simpler parts)





Subtracting the erosion result from the original image

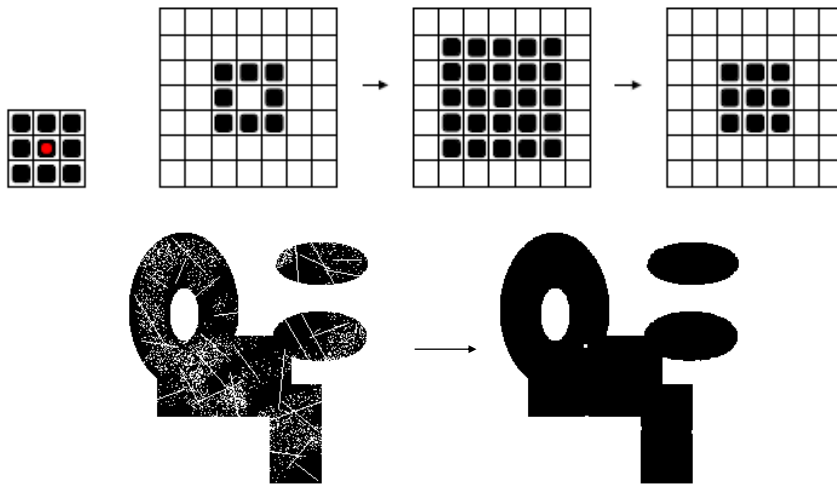
$$\text{Contour} = X - (X \ominus B)$$



Erosion followed by dilation

$$\text{Erosion} = (X \ominus B) \oplus B$$

If the image remains unchanged after opening by the structuring element B , it is open with respect to B



Dilation followed by erosion

$$\text{Closing} = (X \oplus B) \ominus B$$

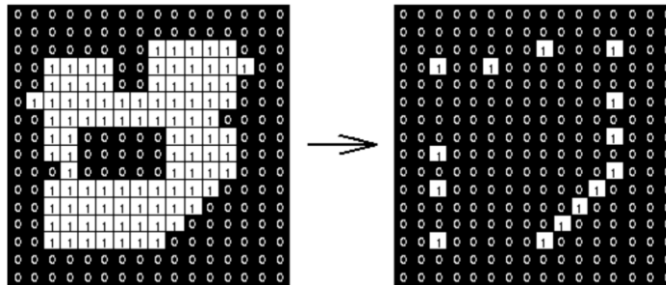
After one opening/closing, the set is already open/closed. Further use of these transformations will not change anything.

	1	
0	1	1
0	0	

	1	
1	1	0
	0	0

	0	0
1	1	0
	1	

0	0	
0	1	1
	1	



$$X \otimes B = \{x : B_1 \subset X \text{ and } B_2 \subset X^c\}$$

$$X \otimes B = (X \ominus B_1) \cap (X^c \ominus B_2) = (X \ominus B_1) \setminus (X \oplus B_2^c)$$

Uses a **composite structuring element**

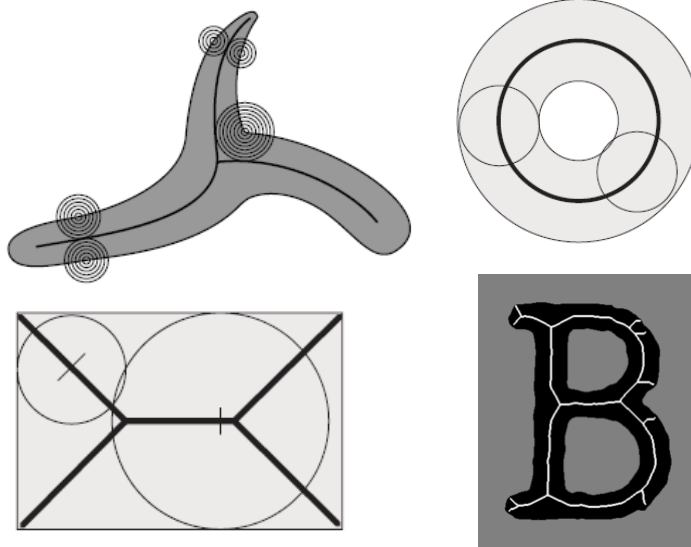
$$B = (B_1, B_2), B_1 \cap B_2 = \emptyset$$

Indicates the match between the composite structuring element and the part of the image

B1 checks the objects and B2 the background

Can be expressed by erosions and dilations

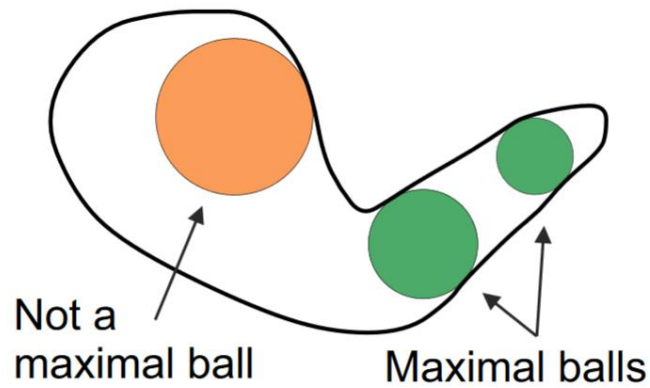
Example: Masks detect four possible configuration of convex corners using hit or miss transformation



It is natural to represent elongated objects by a skeleton
Simplified description of object shape – shape recognition

„Medial Axis Transformation“

A locus of points equidistant from contour (the analogy – grass fire)



Formal definition is based on the concept of the maximal circle:

Circle $B(p, r)$ with center p and radius $r \geq 0$ is the set of points for which distance $d \leq r$

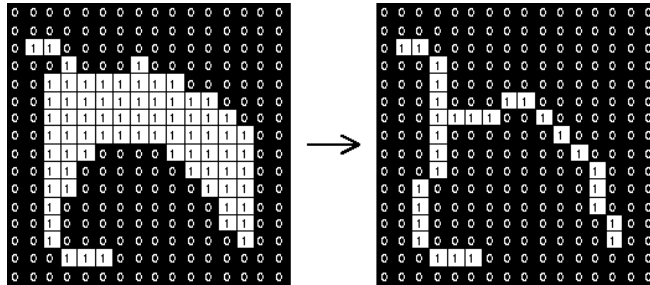
The maximum circle B inscribed into the set X touches its boundary in two or more points

The computational complexity is too high...

- 4x 90° ↻

0	0	0
	1	
1	1	1

	0	0
1	1	0
	1	

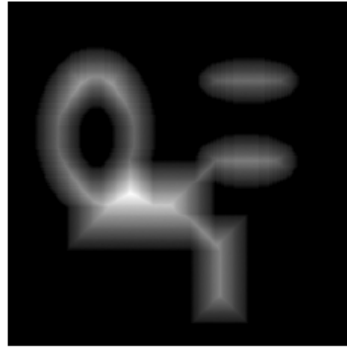
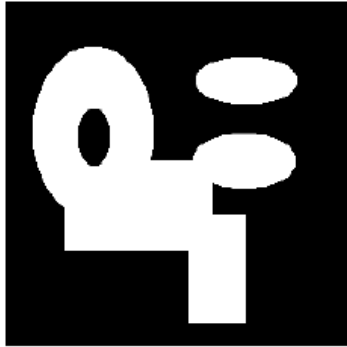


The region is eroded by a suitable structuring element which ensures that the connectivity is not broken

Or by 'hit and miss' transformation ...

- Structuring elements for skeletonization by morphological thinning.
- At each iteration, the image is first thinned by the left hand structuring element, and then by the right hand one,
- and then with the remaining six 90° rotations of the two elements.
- The process is repeated in cyclic fashion until none of the thinnings produces any further change.
- As usual, the origin of the structuring element is at the center.

- Distance transformation of point set A assigns to each point $p \in X$ number – *distance of point p from background A^c*



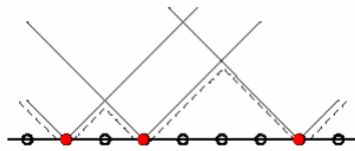
- Two-pass algorithm exists – top-down and bottom-up

Fast and most often used

- Grid formulation, two pass $O(n)$ algorithm for 1D L1 norm
 1. Initialize: For all j
 $D[j] \leftarrow 1P[j]$... indicator function (0 if point, ∞ otherwise)
 2. Forward: For j from 1 up to $n-1$
 $D[j] \leftarrow \min(D[j], D[j-1]+1)$
 3. Backward: For j from $n-2$ down to 0
 $D[j] \leftarrow \min(D[j], D[j+1]+1)$

1 0

0 1



∞	0	∞	0	∞	∞	∞	0	∞
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∞	0	1	0	1	2	3	0	1
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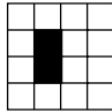
1	0	1	0	1	2	1	0	1
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1D case, L 1 norm: $|x_1 - y_1| + |x_2 - y_2|$

- Grid formulation, **2D L1 norm** (analogous to 1D)
 - Initialize
 - Forward and backward pass
 - Forward pass adds one to closest above and to left, takes min with self
 - Backward pass analogous below and to right

0	1
1	s

s	1
1	0



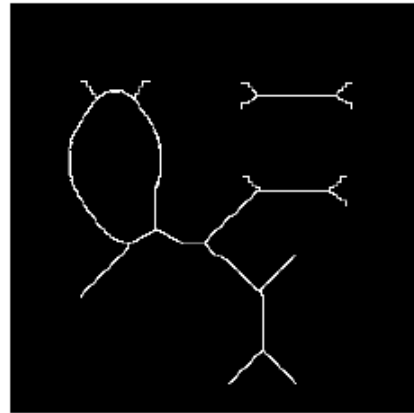
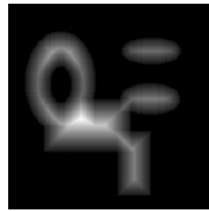
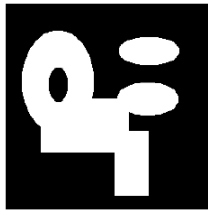
∞	∞	∞	∞
∞	0	∞	∞
∞	0	∞	∞
∞	∞	∞	∞

∞	∞	∞	∞
∞	0	1	∞
∞	0	∞	∞
∞	∞	∞	∞

∞	∞	∞	∞
∞	0	1	2
∞	0	1	2
∞	1	2	3

2	1	2	3
1	0	1	2
1	0	1	2
2	1	2	3

1D case, L1 norm: $|x_1 - y_1| + |x_2 - y_2|$



- Grayscale mathematical morphology is the generalization of binary morphology for images with more gray levels than two
 - Point set $A \in \mathbb{E}^3$
 - The first two components represent 2D image coordinates
 - The third component corresponds to the image value
- The structuring element is a function of two variables.
 - It influences how pixels in the neighborhood of the current pixel are taken into account
 - The value of (**non-flat**) structuring element is added (dilation), resp. subtracted (erosion), when the maximum, resp. minimum is calculated in the neighborhood

- Maximum value of pixels in the neighborhood defined by the structural element:

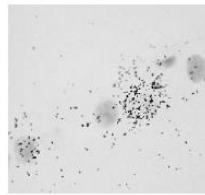
$$\max_{b \in B} f(x + b)$$



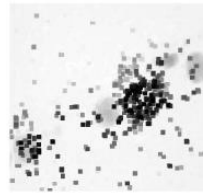
- Minimum value of pixels in the neighborhood defined by the structural element:

$$\min_{b \in B} f(x + b)$$

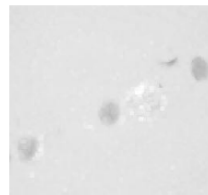




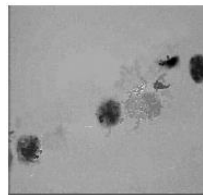
original



erosion



dilation



reconstr. cells